FACULDADE DE ENGENHARIA DA UNIVERSIDADE DO PORTO

ATCS-Automatic Target Classification System

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WORKING VERSION

Mestrado em Engenharia Eletrotécnica e de Computadores

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January 12, 2026

Resumo

A pontuação automática de alvos de papel é um requisito importante no tiro desportivo, particularmente em contextos de treino, onde o feedback em tempo real pode apoiar o desenvolvimento dos atletas. Embora os sistemas de pontuação eletrónica baseados em hardware especializado ofereçam elevada precisão, o seu custo e os requisitos de infraestrutura limitam a sua acessibilidade a clubes amadores. As abordagens baseadas em visão por computador constituem uma alternativa de baixo custo, mas devem satisfazer os elevados requisitos de precisão impostos pelas regras de pontuação decimal da ISSF.

Este trabalho aborda o problema da pontuação automática de alvos em papel da ISSF para as modalidades de carabina de ar comprimido a 10 m e pistola de ar comprimido a 10 m, utilizando sistemas de baixo custo baseados em câmaras. As características físicas dos alvos da ISSF e as respetivas regras de pontuação são analisadas de forma a derivar requisitos de precisão para uma pontuação decimal fiável. Uma revisão do estado da arte mostra que os métodos clássicos de processamento de imagem podem atingir elevada precisão geométrica em condições controladas, mas carecem de robustez, enquanto as abordagens baseadas em deep-learning proporcionam deteção fiável, mas precisão de localização insuficiente quando utilizadas de forma isolada.

Com base nesta análise, é proposta uma arquitetura de sistema híbrida que combina deteção baseada em aprendizagem com refinamento geométrico clássico. É concebido um pipeline modular de processamento de imagem para suportar a deteção robusta do alvo e dos impactos, o tratamento de impactos sobrepostos e a estimativa precisa dos centros, permitindo uma pontuação em conformidade com as regras da ISSF em tempo quase real. A abordagem proposta estabelece uma base para o desenvolvimento e avaliação de um sistema de pontuação automática de baixo custo para alvos em papel da ISSF durante a fase de dissertação.

Abstract

Automatic scoring of paper targets is an important requirement in sport shooting, particularly in training environments where real-time feedback can support athlete development. While electronic scoring systems based on specialized hardware provide high accuracy, their cost and infrastructure requirements limit their accessibility to amateur clubs. Computer-vision-based approaches offer a low-cost alternative but must satisfy the high accuracy requirements imposed by ISSF decimal scoring rules.

This work addresses the problem of automatic scoring of ISSF 10 m air rifle and 10 m air pistol paper targets using low-cost camera-based systems. The physical characteristics of ISSF targets and the scoring rules are analyzed to derive accuracy requirements for reliable decimal scoring. A review of the state of the art shows that classical image-processing methods can achieve high geometric precision under controlled conditions but lack robustness, while deep learning approaches provide reliable detection but insufficient localization accuracy when used in isolation.

Based on this analysis, a hybrid system architecture is proposed that combines learning-based detection with classical geometric refinement. A modular image-processing pipeline is designed to support robust target and bullet-hole detection, overlap handling and precise center estimation, enabling ISSF-compliant scoring in near real time. The proposed approach establishes a foundation for the development and evaluation of a low-cost automatic scoring system for ISSF paper targets during the dissertation phase.

UN Sustainable Development Goals

The United Nations Sustainable Development Goals (SDGs) define a global framework for promoting sustainable development through social, economic, and technological progress. Although this thesis is not directly focused on addressing a specific SDG, it contributes indirectly by developing a low-cost technological solution that supports accessibility and innovation in sports training environments.

In particular, this work is loosely aligned with the following goal:

SDG 9 – Industry, Innovation and Infrastructure: The development of a low-cost, computer-vision-based automatic scoring system demonstrates the application of digital technologies and innovation to improve existing infrastructures in sports facilities, making advanced training tools more accessible to amateur clubs and individual athletes.

Acknowledgements

I would like to express my gratitude to my supervisor for the constant availability to discuss ideas and for granting me access to the local shooting center, where I was able to start testing some parts of the system and gain practical experience by shooting at targets myself.

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SDG	Sustainable Development Goal
CNN	Convolutional Neural Network
ISSF	International Shooting Sports Federation

Chapter 1

Introduction

Sport shooting requires accurate and reliable scoring of targets in order to evaluate athlete performance during training and competition, as well as to improve the training experience. Traditionally, paper targets have been scored manually, a process that is time-consuming, error-prone and unsuitable for real-time feedback. To overcome these limitations, a variety of automatic scoring systems have been developed over the years, including solutions based on acoustic triangulation, laser beam interruption and electronic impact sensing. While these systems offer high accuracy and reliability, they typically involve specialized hardware and high cost, making them less accessible to amateur clubs and training environments.

Computer-vision-based scoring systems provide a low-cost alternative by using standard imaging hardware and image-processing algorithms to detect and score bullet impacts on paper targets. Advances in image processing and machine learning have increased the feasibility of such approaches, enabling automatic detection of bullet holes under challenging conditions. However, the application of computer vision to ISSF-compliant paper targets remains challenging due to the small dimensions involved, the use of decimal scoring and the high accuracy requirements imposed by official rules.

This work focuses on the automatic scoring of ISSF 10 m air rifle and 10 m air pistol paper targets using low-cost camera-based systems. The objective is to design a computer-vision-based scoring system capable of detecting bullet holes, estimating their positions with sufficient accuracy and computing ISSF-compliant decimal scores in near real time. Particular emphasis is placed on achieving a balance between robustness to real-world imaging conditions and the geometric precision required for accurate scoring.

To address this problem, the dissertation explores a hybrid system architecture that combines learning-based methods and classical image-processing techniques. Convolutional neural networks are considered for robust target and bullet-hole detection, while geometric refinement methods are employed to recover accurate center coordinates required for scoring. The system is designed to operate in controlled indoor shooting ranges using low-cost hardware, without relying on specialized electronic targets.

The remainder of this document is organized as follows. Chapter 2 analyzes the physical and regulatory constraints of ISSF paper targets and reviews the state of the art in computer-vision-based target scoring systems. Chapter 3 presents the proposed system architecture and outlines the image-processing methods considered for target localization, bullet-hole detection and geometric refinement. Chapter 4 concludes the report with a summary of the main findings and a discussion of the expected contributions and future work during the dissertation phase.

Chapter 2

Problem Definition and State of the Art

2.1 Introduction

Scoring targets is an essential task in sport shooting, and a variety of systems have been developed over the years to replace the time-consuming manual scoring. Common technologies include triangulation of acoustic signals, laser-beam interruption and impact sensing. These technologies represent high-accuracy but high-cost solutions, and are well-suited for professional events. Many amateur sport-shooting clubs are interested in low-cost, real-time scoring systems that do not require the same level of precision. Computer-vision-based systems offer a low-cost alternative capable of detecting and scoring paper targets.

This section reviews the state of the art in computer-vision-based target scoring systems. This includes classical image processing methods, and more recent deep learning approaches.

2.2 Physical and Regulatory Requirements of ISSF Paper Targets

ISSF defines the dimensions and scoring rules for paper targets used in 10 m air rifle and 10 m air pistol events.

Table 2.1: ISSF 10 m Air Rifle target ring diameters [3]

Ring	Diameter (mm)
10	0.5
9	5.5
8	10.5
7	15.5
6	20.5
5	25.5
4	30.5
3	35.5
2	40.5
1	45.5

Table 2.2: ISSF 10 m Air Pistol target ring diameters [3]

Ring	Diameter (mm)
10	11.5
9	27.5
8	43.5
7	59.5
6	75.5
5	91.5
4	107.5
3	123.5
2	139.5
1	155.5

Figure 2.1: ISSF 10 m Air Rifle target [3]

Figure 2.2: ISSF 10 m Air Pistol target [3]

Figure 2.1 illustrates the ISSF 10 m air rifle paper target, while Fig. 2.2 shows the corresponding 10 m air pistol target. The official ring diameters for both targets are listed in Table 2.1 and Table 2.2, respectively.

For the 10 m air rifle paper target, the innermost ring (ring 10) is reduced to a small white dot at the center of the target with a diameter of 0.5 mm, as shown in Fig. 2.1. The ring diameters increase arithmetically by 5 mm up to the outermost ring (ring 1), which has a diameter of 45.5 mm, as summarized in Table 2.1. The inner black region extends from ring 10 to ring 4, with a total diameter of 30.5 mm.

The 10 m air pistol paper target is larger, as shown in Fig. 2.2. The innermost ring (ring 10) has a diameter of 11.5 mm, while the outermost ring (ring 1) has a diameter of 155.5 mm. The ring diameters increase arithmetically by 16 mm, as listed in Table 2.2. The inner black region extends from ring 10 to ring 7, with a diameter of 59.5 mm.

The dimensions summarized in Fig. 2.1 and Table 2.1 highlight the extremely small scale of the air rifle target, which directly motivates the accuracy analysis presented in the following section.

The continuous radial scoring model can be written as

$$s_c(r) = 11.0 - k\,r, \tag{2.1}$$

where r is the radial distance between the shot center and the target center, and k is a scale factor and corresponds to the inverse radius increment

$$k = \frac{2}{\Delta d_{1.0}} \tag{2.2}$$

between adjacent rings.

ISSF decimal scoring applies rounding and clipping to the continuous scoring model with the formula

$$s(r) = \mathrm{clip}\left(\frac{\lfloor 10 s_c(r) \rfloor}{10}, 0, 10.9\right), \tag{2.3}$$

where $\lfloor\cdot\rfloor$ denotes the floor operator and $\mathrm{clip}(x,a,b) = \min(\max(x,a),b)$ limits the score to the valid ISSF range.

2.3 Accuracy Requirements

The scoring scale resolution and the smallest paper target size impose accuracy requirements for any scoring system. Due to the small dimension of the 10m-air-rifle paper target, the required accuracy goes into the sub milimeter range.

To understand the accuracy requirements, we analyse the measurements relative to one interval of constant score. The length of the interval is given by

$$\Delta r_{0.1} = 0.05 \cdot \Delta d_{1.0} \tag{2.4}$$

and corresponds to the increment in radial distance that produces a change of 0.1 in the continuous score.

The measurement error is modeled as a uniform random variable in the interval $[-E, E]$. Let R denote the ground-truth radial position within a decimal scoring interval and let ε denote the measurement error. The measured value is $M = R + \varepsilon$. Assuming R and ε are independent, the probability density function (pdf) of M is given by the convolution of the pdfs of R and ε, as illustrated in Fig. 2.3.

Regardless of how small the error is, there will always be a non-zero probability for incorrect scoring by 0.1. The probability of correctly scoring a bullet is given by the integral of the pdf of the measurement within the scoring interval, which is

$$P = 1 - \frac{E}{2 * \Delta r_{0.1}} \tag{2.5}$$

.

Solving for the accuracy E, we obtain:

$$E = 2\Delta r_{0.1}\,(1 - P) \tag{2.6}$$

.

Table 2.3 shows that achieving high probabilities of correct decimal scoring requires increasingly strict bounds on the radial measurement error. For the 10 m air rifle target, a probability of 90% already requires a measurement accuracy better than 0.05 mm, while probabilities above 95% demand accuracies below 0.03 mm. These values illustrate the high accuracy requirements imposed by ISSF decimal scoring and motivate the use of precise geometric localization methods in automatic paper target scoring systems.

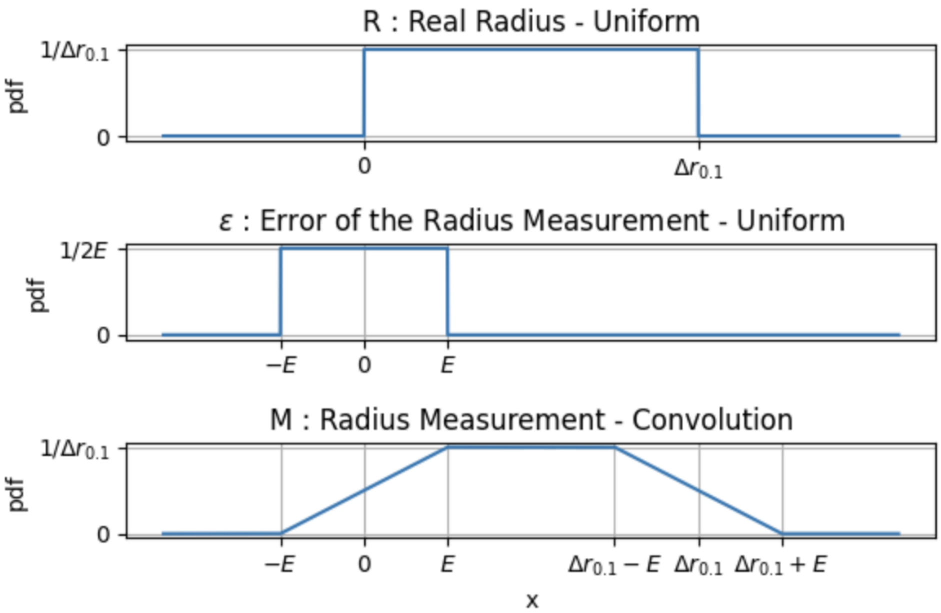

Figure 2.3: Bounded-error model: uniform ground-truth position within a scoring interval and uniform measurement error, and the resulting measurement pdf obtained by convolution.

Table 2.3: Required measurement accuracy E as a function of the probability P of correct decimal scoring for a 10 m air rifle target ($\Delta r_{0.1} = 0.25$ mm).

Probability P **(%)**	**Required accuracy** E **(mm)**
80	0.10
85	0.075
90	0.050
95	0.025
99	0.005

2.4 Classical Computer Vision Approaches

Classical computer vision approaches to paper target scoring typically rely on explicit image segmentation and geometric reasoning. In contrast to learning-based methods, they aim to recover physically meaningful quantities—such as the target center, ring radii and bullet-hole centers—and then compute the score by applying the official scoring rules. These pipelines can achieve high localization accuracy under controlled imaging conditions, but they often become fragile when lighting varies, the background is cluttered, or bullet holes overlap.

2.4.1 Geometric scoring with perspective correction and contour-based circle fitting

A representative geometric pipeline is proposed by Yu *et al.* [6], who score paper targets using only classical image processing and circle fitting. The method first computes a projective transformation to obtain a fronto-parallel view of the target, compensating for camera viewpoint. After perspective correction, the outer contour of the target is extracted and a circle is fitted to estimate the target center and scale. Bullet holes are then isolated via thresholding and morphological operations, and connected components are extracted. For each bullet-hole region, contours are computed and a circle model (e.g., minimum enclosing circle or least-squares circle fitting) is fitted to estimate the bullet-hole center. Finally, the score is computed from the distance between the bullet-hole center and the target center using the known ISSF ring geometry.

The main strength of this approach is that it reconstructs the target geometry and estimates centers via contour-based fitting, which is well-aligned with ISSF-compliant scoring. However, it assumes that segmentation is reliable, which is typically achieved under controlled backgrounds and stable lighting. As a result, performance degrades when bullet holes have bad contrast against the paper, lighting changes, or when bullet holes overlap, since the connected components no longer correspond to individual impacts.

2.4.2 Segmentation of overlapping bullet holes using watershed and circle fitting

Overlapping impacts are a major failure mode for connected-component-based pipelines. Liu *et al.* [5] address this scenario using an improved watershed segmentation strategy. Their pipeline applies grayscale preprocessing and morphological operations, followed by area-based filtering to suppress non-hole regions such as ring markings and text. Overlapping bullet holes are then separated using a distance-transform-driven watershed method, and boundaries are extracted. For each separated region, least-squares circle fitting is used to estimate individual hole centers and radii.

This work demonstrates that, when the preprocessing is successful, classical segmentation can separate strongly overlapping holes and still recover geometrically meaningful centers. Nonetheless, the method depends on threshold choices and assumptions about bullet-hole appearance (e.g., expected region areas). In addition, the paper focuses on cropped regions around overlapping holes rather than full target images, so it does not fully address end-to-end robustness in a complete target scoring system.

2.4.3 Temporal differencing and Hough-based circle detection for live scoring

A complementary classical strategy for live scoring is to exploit temporal information. Li *et al.* [4] propose a method that processes pairs of images captured before and after each shot. After geometric correction (homography), the pre-shot image is subtracted from the post-shot image, only keeping the pixels that changed. The pipeline then applies binarization, morphological cleaning and edge detection, followed by a generalized Hough transform to detect the circular bullet-hole contour and estimate its center. A Hough transform can also be used to localize a reference ring or the target center, enabling score computation from the measured radial distance.

The main advantage of temporal differencing is facilitating the segmentation task. However, it requires a stable camera and assumes that exactly one new shot occurs between consecutive frames; camera motion or multiple new impacts can break the subtraction assumption. Despite these constraints, temporal differencing is attractive as an optional component in real-time systems where images are captured continuously and the shot timing is known.

2.4.4 Discussion

Overall, classical methods are attractive because they compute scores from explicit target geometry and can be computationally lightweight. Their limitations arise primarily from the sensitivity of segmentation to illumination changes, background clutter and target mounting frames (like a metal frame). In addition, classical pipelines are typically parameter-heavy, relying on multiple thresholds and heuristic parameters (e.g., intensity thresholds, morphological kernel sizes and area constraints) that must be carefully tuned for each imaging setup. These observations motivate hybrid designs that preserve geometric scoring and center estimation, while improving robustness through stronger detection mechanisms (e.g., learning-based detectors or temporal differencing when available).

2.5 Deep Learning-Based Approaches

Recent work has explored the use of deep learning, particularly convolutional neural networks (CNNs), for automatic detection of bullet holes on paper targets. These approaches typically frame the problem as an object detection or instance segmentation task, with the goal of localizing bullet holes under challenging imaging conditions.

A representative study is presented by Butt *et al.* [2], who apply the YOLOv8 object detection framework to detect bullet holes on indoor shooting targets. The authors report high detection performance, with mean Average Precision (mAP) values exceeding 95% under controlled indoor conditions. A key advantage of the YOLOv8-based approach is its computational efficiency: inference times are on the order of a few milliseconds per image, making the method suitable for real-time or near-real-time applications. The study demonstrates that CNN-based detectors can robustly identify bullet holes despite moderate lighting variations and background clutter, outperforming classical threshold-based methods in detection reliability.

However, the detected bullet holes are represented by bounding boxes, and the paper does not address sub-region analysis or precise center estimation beyond the detector output. Consequently, while detection accuracy is high, the geometric precision required for ISSF-compliant decimal scoring is not directly achieved.

A related line of work is presented by Ahmed *et al.* [1], who employ a Mask R-CNN architecture with a ResNet-50 backbone to detect and segment bullet holes on outdoor shooting targets. This work demonstrates the robustness of deep learning methods in more adverse conditions, including variable illumination, weather effects and non-uniform backgrounds. The use of instance segmentation allows the network to recover the approximate shape of bullet holes rather than only bounding boxes, which is advantageous for visual interpretation and robustness.

Despite this increased robustness, the reported localization accuracy remains limited by the resolution of the segmentation masks and the network output grid. The study focuses primarily on detection and segmentation performance metrics and does not explicitly analyze the accuracy of the estimated bullet-hole centers relative to the precision required for ISSF decimal scoring. In addition, overlapping bullet holes are not explicitly addressed, and the segmentation results are evaluated mainly on isolated impacts.

Overall, deep learning approaches offer significant advantages in terms of robustness to illumination changes, background clutter and target wear, and they enable fast, real-time detection on low-cost hardware. However, accurately estimating bullet-hole center coordinates with submillimeter precision remains a major challenge. The detector outputs (bounding boxes or coarse masks) are not sufficiently precise for high-accuracy scoring on their own, and none of the reviewed deep learning studies explicitly address the problem of overlapping bullet holes. These limitations motivate hybrid approaches that combine CNN-based detection for robustness with classical geometric refinement for precision.

2.6 Discussion and Summary

This chapter reviewed the physical and regulatory constraints imposed by ISSF paper targets and analyzed their implications for automatic scoring systems. Due to the small dimensions of the 10 m air rifle target and the use of decimal scoring, even small localization errors can lead to incorrect scores. A simple bounded-error model was introduced to illustrate how scoring accuracy depends directly on the measurement error, highlighting the high accuracy requirements imposed by ISSF rules.

The state of the art in computer-vision-based target scoring was then reviewed. Classical image processing approaches rely on geometric modeling and can achieve high localization accuracy under controlled conditions. However, these methods are sensitive to illumination changes, background clutter and overlapping bullet holes, and they often require careful tuning of multiple heuristic parameters. In contrast, deep learning-based approaches demonstrate strong robustness and real-time performance, particularly for bullet-hole detection, but their outputs are typically limited to bounding boxes or coarse segmentation masks (Mask R-CNN), which do not provide the geometric precision required for ISSF-compliant decimal scoring.

Overall, the literature shows that no single approach fully satisfies both the robustness and precision requirements of automatic ISSF paper target scoring. These observations motivate a hybrid system design that combines the robustness of learning-based detection with the precision of classical geometric refinement. The following chapter presents the proposed system architecture based on this hybrid approach.

Chapter 3

Proposed System Architecture

3.1 System Overview

The proposed system is designed to provide automatic scoring of ISSF paper targets using a low-cost USB camera and computer vision techniques. In addition to shot detection and scoring, the system supports data storage and visualization of a virtual target to enable the management of athletes, training sessions and competitions.

At a high level, the system consists of five main components: image acquisition, image processing, scoring logic, data management and visualization. A fixed USB camera captures images of the paper target after each shot and transmits them to a personal computer for processing. The image-processing subsystem analyzes the acquired images to localize the target, detect bullet holes and estimate their geometric properties. Based on these measurements, the scoring module computes ISSF-compliant decimal scores. The resulting coordinates and scores are stored in a database, where they are associated with athletes, sessions and competitions, and can be visualized through a virtual target representation.

Figure 3.1 illustrates the high-level architecture of the proposed system and the main processing stages from image acquisition to target visualization.

While the complete system provides end-to-end functionality for automatic target scoring, the primary focus of this thesis is the image-processing subsystem. In particular, the research addresses the problem of robust and accurate bullet-hole localization under the constraints identified in Chapter 2. The remaining components, including data storage and visualization, serve to integrate the image-processing results into a usable scoring system but are not the main object of investigation.

The image-processing subsystem is designed as a modular pipeline, allowing different detection and refinement strategies to be evaluated and compared. Robust bullet-hole detection is treated separately from precise geometric refinement, enabling the system to combine complementary approaches and to merge robustness and accuracy. The following sections describe the design of this subsystem in detail, beginning with image acquisition assumptions and progressing through detection, refinement and scoring.

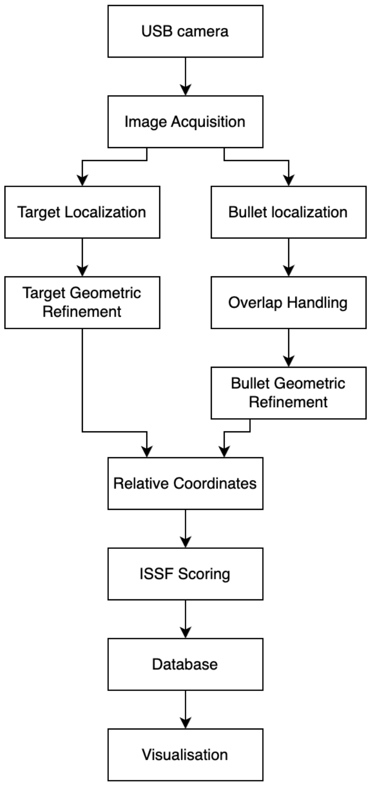

Figure 3.1: High-level architecture of the proposed system.

3.2 Image Acquisition and Operating Assumptions

Images of the paper target are acquired using a fixed USB camera connected to a personal computer. The camera is mounted at a known and approximately constant distance from the target and remains stationary during operation. This fixed setup allows the use of geometric constraints and calibration procedures in later processing stages.

Figure 3.2 shows that the camera will be placed at an angle, looking at the targets from the side of the shooting line.

The system is intended to operate in an indoor shooting range, where lighting conditions can be controlled to a certain extent but are not assumed to be perfectly uniform. Variations in illumination, shadows and reflections may still occur. No additional white background board is assumed behind the target by default.

However, the operating environment allows for limited control of background appearance if required. In particular, the use of a uniform background color behind the target can be introduced to make segmentation easier and more robust. This option provides additional flexibility while remaining consistent with the low-cost and controlled indoor deployment scenario considered in this work.

Images are captured at a sufficient resolution to allow reliable detection of bullet holes and target features. Image acquisition is triggered after each shot, either manually or through an external

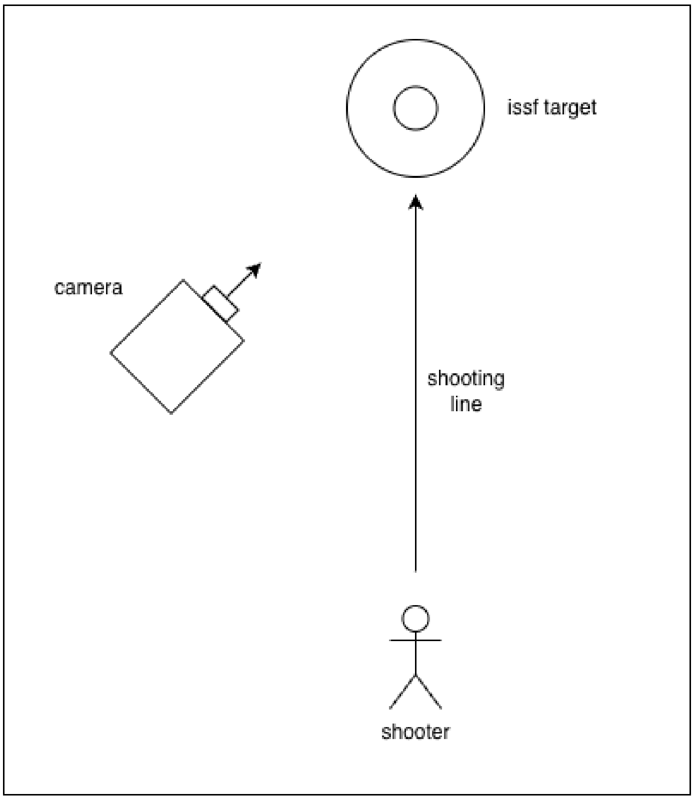

Figure 3.2: Camera placement relative to the shooting line and paper target.

signal, such that each image contains at most one new bullet hole relative to the previous state of the target.

These assumptions reflect a realistic low-cost deployment scenario and are consistent with the constraints discussed in Chapter 2. They guide the design of the image-processing pipeline described in the following sections.

3.3 Image-Processing Subsystem and Considered Methods

The image-processing subsystem is structured as a sequence of modular processing stages, as illustrated in Fig. 3.1. For each stage, different algorithmic strategies are considered based on the trade-off between robustness, accuracy and computational complexity. This section outlines the main methods.

Target Localization

Accurate localization of the target center is essential, as all subsequent measurements are defined relative to this reference. Two complementary strategies are considered. A learning-based approach may be used to detect the target or the central black region using a CNN trained on target images captured under different conditions, including targets with existing bullet holes. Alternatively, classical image-processing methods based on intensity thresholding and geometric constraints may be applied to segment the black region of the target and estimate its center. In both cases, the initial detection defines a region of interest that can be further refined using classical methods operating on a cropped and normalized image.

Bullet-Hole Detection (Localization)

Bullet-hole detection localizes all bullet holes within the target region. Learning-based object detection methods, including both pretrained detectors (e.g., YOLO-style or region-based CNNs) and custom lightweight CNNs, are considered for this stage. To reduce data collection effort and improve generalization, synthetic data generation may be explored to pretrain networks on circular shapes resembling bullet holes under varying contrast and noise conditions.

As an alternative or complement to learning-based detection, classical approaches based on intensity thresholding and morphological operations may be employed, particularly in scenarios where background appearance can be controlled. Under controlled conditions, these methods are very robust.

Overlap Handling and Temporal Information

In cases where multiple bullet holes are present in close proximity, a single detected region may contain more than one impact. To address this situation, segmentation-based techniques such as watershed algorithms driven by distance transforms are considered for separating overlapping bullet holes. The number of impacts may be estimated to guide the application of such methods.

When images are acquired sequentially during live shooting, temporal information can also be used. Image subtraction between consecutive frames may be used to isolate newly created bullet holes. In this approach, the camera movement must be approximately zero and shooting timing must be detected.

Geometric Refinement and Center Estimation

Following coarse detection, a geometric refinement stage is applied to estimate precise bullet-hole properties, including center coordinates and hole radius. Classical contour-based methods are considered for this purpose, relying on thresholding, edge detection and contour extraction within a cropped region of interest. Circle fitting techniques can then be applied to the extracted contours to estimate the bullet-hole center.

As an alternative, Hough transform-based circle detection may be used to directly estimate circle parameters from edge maps. Learning-based refinement approaches, such as small regression or segmentation networks operating on localized regions, are also considered as potential alternatives to purely classical methods.

Scoring Integration

The outputs of the target localization and geometric refinement stages are combined to compute radial distances between bullet-hole centers and the target center. These distances are then converted into decimal scores according to the ISSF scoring rules described in Chapter 2. The resulting shot coordinates and scores are forwarded to the scoring, data management and visualization components of the system, and a virtual target representation is generated.

Chapter 4

Final Considerations

This work addressed the problem of automatic scoring of ISSF paper shooting targets using low-cost computer vision techniques. The motivation for the study arises from the need for accessible scoring systems in amateur and training environments, where existing electronic solutions are often too expensive or require specialized infrastructure.

An analysis of the physical characteristics and scoring rules of ISSF paper targets showed that decimal scoring imposes high accuracy requirements, particularly for 10 m air rifle targets. A simple bounded-error model was introduced to illustrate how measurement accuracy directly affects the probability of correct scoring, highlighting that even small localization errors can result in score deviations. This analysis established the fundamental constraints that any automatic scoring system must satisfy, independently of the specific algorithms employed.

The state of the art in computer-vision-based target scoring was reviewed, covering both classical image-processing methods and recent deep learning approaches. Classical methods rely on explicit geometric modeling and can achieve high precision under controlled conditions, but they are sensitive to illumination changes, background clutter and overlapping bullet holes, and often require careful parameter tuning. Deep learning-based approaches are more robust and have low inference times, but the precision of their outputs is typically insufficient for high-accuracy ISSF scoring when used alone. The literature review indicates that neither class of methods alone fully addresses the combined requirements of robustness and precision.

Based on these observations, a hybrid system architecture was proposed. The system combines robust detection mechanisms, potentially based on CNNs, with geometric refinement stages designed to recover accurate bullet-hole and target center coordinates. A modular image-processing pipeline was outlined, allowing different strategies to be evaluated for target localization, bullet-hole detection, overlap handling and geometric refinement.

The proposed architecture defines a clear structure for the dissertation phase, during which the individual modules will be implemented, evaluated and integrated into a complete scoring system. Future work will focus on selecting and refining for each module, assessing their performance under realistic operating conditions, and quantifying the scoring accuracy using both synthetic and real target data.

Overall, this work establishes a solid foundation for the development of a low-cost, computer-vision-based automatic scoring system for ISSF paper targets, and clearly defines the technical challenges and methodological directions to be addressed in the subsequent implementation and evaluation stages.

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